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NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA-136119

B. Tech. (PIE) – 3rd Sem.; Mid-SEM Exam –I

September-2025

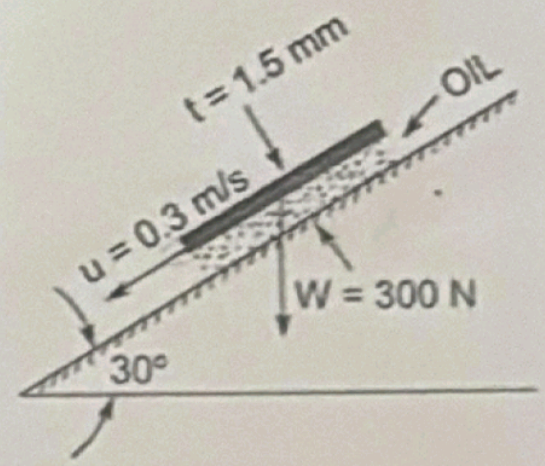
PIPC - 202 | **Fluid Mechanics & Machines**

Time: 50 Minutes | *Max. Marks: 15*

Note: Attempt all questions. Assume suitable missing data, if any.

Q-1: Calculate the dynamic viscosity of an oil, which is used for lubrication between a square plate of size 0.8 m × 0.8 and an inclined plane with angle of inclination 30∘ as shown in Figure. The weight of the square plate is 300 N and it slides down the inclined plane with a uniform velocity of 0.3 m/s. The thickness of oil film is 1.5 mm.

(3)



Q-2: The inlet and throat diameters of a horizontal venturimeter are 30 cm and 10 cm respectively. The liquid flowing through the meter is water. The pressure intensity at inlet is 13.734 N/cm² while the vacuum pressure head at the throat is 37 cm of mercury. Find the rate of flow. Assume that 4% of the differential head is lost between the inlet and throat. Find also the value of C_d for the venturimeter.

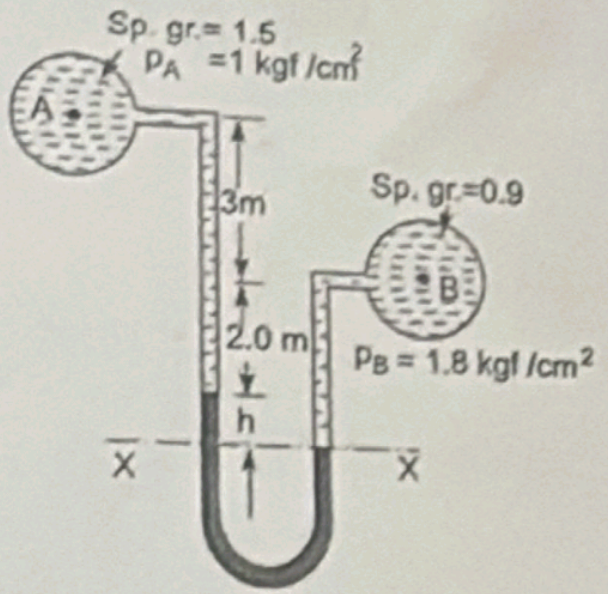
(4)

Q-3: Derive an expression for the capillary rise of a liquid in the vertical annular space between two concentric cylinders of inner radius R_i and outer radius 'R_o'. Assume the liquid has surface tension σ, contact angle θ, density ρ, and is under gravity g. Show that the rise depends only on the gap between the cylinders.

(3)

Q-4: A differential manometer is connected at the two points A and B of two pipes as shown in Figure. The pipe A contains a liquid of sp. gr. = 1.5 while pipe B contains a liquid of sp. gr. = 0.9. The pressures at A and B are 1 kgf/cm² and 1.80 kgf/cm² respectively. Find the difference in mercury level in the differential manometer.

(3)



Q-5: With neat sketches, explain the conditions of equilibrium for floating bodies.

(2)